

Hydrocarbon Potential – Russia, Central Asia, Falklands

Executive Summary: Russia holds the world's largest conventional hydrocarbon endowment, with officially disclosed oil reserves of ≈ 17.8 bn tonnes (≈ 130 bn bbl) and gas reserves ≈ 48.8 tcm 【9†L57-L63】. Its main producing province is the West Siberian Basin (major oil/gas source: the Bazhenov shale) 【54†L213-L221】. In **Central Asia**, Kazakhstan (oil ≈ 30 bn bbl, gas ≈ 85 Tcf) 【104†L147-L154】 dominates; major Kazakh fields are Tengiz, Kashagan and Karachaganak. Turkmenistan has the world's fifth-largest gas reserves (~ 400 Tcf) and modest oil (0.6 bn bbl) 【104†L156-L164】. Uzbekistan's hydrocarbon base is smaller (≈ 0.6 bn bbl oil, 65 Tcf gas) 【104†L168-L174】 , concentrated in the Bukhara-Khiva, Fergana and Ustyurt basins. **Falklands:** the offshore North Falkland (Sea Lion) basin contains ~ 0.9 bn bbl recoverable oil 【75†L141-L149】. Geologically, the Caspian/Central Asian plays range from Paleozoic reef carbonates (e.g. Tengiz Permian, Karachaganak Carboniferous) to Mesozoic clastics (Kashagan Carboniferous, Falklands Cretaceous sands) 【59†L157-L161】 【64†L163-L171】. These yield a mix of heavy sour oil (e.g. Tengiz $\sim 17\%$ sulfur 【62†L24-L32】) and wet gas/condensate (Karachaganak).

【97†embed_image】 *Figure: Major oil (red) and gas (blue) pipeline networks in Russia and Central Asia (EIA).* This infrastructure illustrates export routes (e.g. Druzhba, CPC pipeline) and fields (stars).

Key Basins/Fields (Reserves, Risk, Cost):

Field/Basin	Country	Type	Reserves (oil/gas)	Break-even (\$/bbl)	Hazards & Risk
Samotlor (West Siberia)	Russia	Oil	Oil in place ~ 4 bn bbl 【57†L155-L162】 (peak giant; ~ 2.6 Gt produced)	$\approx \$15-20$	Mature heavy field, extreme cold, permafrost, pipeline age.
Urengoy (Yamal)	Russia	Gas	~ 10 tcm gas (supergiant)	$\approx \$4-6/\text{Mcf}$	Arctic operations, long-distance pipelines (e.g. Yamal-Europe/ NordStream)

Field/Basin	Country	Type	Reserves (oil/gas)	Break-even (\$/bbl)	Hazards & Risk
Tengiz (Pre-Caspian)	Kazakhstan	Oil/Gas	Oil in place 25.5 bn bbl 【61†L160-L162】 ; large associated gas	≈\$15–25	Very sour (~17% S 【62†L24-L32】), deep carbonate reservoir (Gulf salt overburden), requires high-cost sulfur removal.
Kashagan (Caspian)	Kazakhstan	Oil	Recoverable ~13 bn bbl 【59†L169-L172】 (field total ~38 bn bbl)	≈\$40–60	Offshore, H ₂ S-rich (sour), very high capex (multi-100 M\$ wells, offshore construction), heavy environmental regs.
Karachaganak (Pre-Caspian)	Kazakhstan	Gas/Condensate	≈1.2 tcm gas; ~1 Gt liquids (oil+condensate) 【64†L163-L171】	≈\$10–15	Retrograde gas-condensate (Permian reef), sour gas (3–5% H ₂ S 【64†L203-L211】), complex reservoir management (pressure maintenance).
Sea Lion (North Falkland)	Falklands	Oil	~917 mm bbl recoverable 【75†L141-L149】 (300 m API field)	≈\$50–70	Remote offshore: deepwater storms, 8,000+ km to markets, geopolitical dispute with Argentina, strict UK environmental standards.

Russia

Reserves: Official figures (Rosnedra) give **oil** ABC1 (“proved”) ≈17.8 bn t (≈130 bn bbl) and gas ABC1 ≈48.8 tcm 【9†L57-L63】 . Probable (C2) adds ≈10.9 bn t oil and 19.6 tcm gas 【9†L57-L63】 (others cite ≈85–130 Tcf). Russia's *BP Statistical Review* (2025) and **Energy Dept.** report roughly 80–100 bn bbl oil and

~1,700 Tcf gas. Key fields: *Samotlor* (largest oil), *Priobskoe*, *Prirodnoe*; gas *Urengoy*, *Yamburg*, *Medvezhye*, and new *Yamal* fields. Eastern Russia: *Sakhalin I/II* (Liquified gas and oil in Far East).

Basins/Plays: The giant **West Siberian Basin** (Jurassic-Cretaceous clastic reservoirs) contains ~144 bn bbl oil and ~1,300 Tcf gas discovered [54†L233-L238] . The Bazhenov Formation (Upper Jurassic shale) is the prolific source rock, generating >80% of field oil [54†L213-L221] . Other plays: Paleozoic carbonate buildups (Pre-Caspian: Tengiz's Carboniferous carbonates under Permian salt [70†L378-L387] , Karachaganak Permian/Carboniferous reefs [64†L163-L171]); and Eastern Siberia's Bazhenov analogs. *Kolguev-Kara* shelf (oil/gas prospects). The Far East (Sakhalin) is Miocene sandstones. Reservoir quality ranges from tight (tight Bazhenov) to high-quality turbidites (Tyumen, Neocomian).

Drilling/Development: Russia's onshore fields are largely mature. Best practice includes horizontal drilling and water/gas injection (e.g. gas reinjection at Tengiz) to maintain pressure. *Mature fields* like Samotlor use infill and EOR (Polymer, CO₂ injection). *New developments* focus on frontier basins: Arctic (Laptev, Barents) – very high cost rigs; East Siberia (Komi, Yakutia). The **Yamal** project is being drilled horizontally and tied-in (e.g. 2010s drilling at Bovanenkovo). Shallow onshore is conventional drilling (\$5–10 M per well). In contrast, *Arctic deepwater* (Karachan, Rossenberg) requires specialized ice-class rigs or platforms.

Hazards/Regulation: Climate and geography are severe: Arctic freeze, permafrost (complicating wells/pipe anchors), remote logistics. The Caspian/Pre-Caspian has moderate seismicity (some earthquakes near Turkmen-Kaz border). Politically, sanctions (UK/US/EU) severely restrict imports of Western drilling technology and investment since 2022. Regulatory oversight is by **Rosprirodnadzor/Rosnedra** (permit approvals are strict; local content rules increasing). Foreign firms generally require JV with local companies; projects in Border regions (Arctic shelf) face new "Sovereignty" laws.

Logistics & Exports: Russia's oil export routes: the **Druzhba** pipeline (Western Europe, ~65 Mtpa), **ESPO** (east to Pacific ports), **CPC** (Caspian Pipeline to Black Sea), and Baltic terminals (Primorsk/Ust-Luga tanker loading). Gas exports via pipeline: **Yamal-Europe** (through Poland), **Nord Stream 1/2** (to Germany via Baltic), **TurkStream** (via Turkey), and Ukraine routes (now minimal). LNG: Arctic LNG-2 (Sabetta) exports to Asia. For UK/Europe, Russian exports travel via pipeline or tankers through the Baltic/Mediterranean. Key chokepoints: **Strait of Bosphorus** (for CPC tankers), **Capitals** (finance) for sanctions enforcement. UK gas purchases (pre-2022) were via Nord Stream; now UK relies on LNG.

flowchart LR

subgraph Russia

R_Oil[RS Oil fields] --> Druzhba[Druzhba pipeline (to Europe)]

R_Oil --> ESPO[ESPO pipeline (to Pacific)]

R_Gas[RS Gas fields] --> NordStream[Nord Stream (to Germany)]

R_Gas --> TurkStream[TurkStream (to Turkey/EU)]

end

subgraph Central_Asia

K_Oil[Kazakh Oilfields] --> CPC[CPC pipeline (to Black Sea)]

CPC --> Novoro[Novorossiysk terminal] --> Med[Ship via Med/Gibraltar]

K_Oil --> CNPCo[Kazak-China oil pipeline]

K_Gas[Kazakh Gasfields] --> CNPCg[Central Asia-China gas pipeline]

K_Gas --> CA_Center[Central Asia – Russia gas pipeline] --> Russia

T_Gas[Turkmen Gasfields] --> China_Gas[China gas pipeline]

T_Gas --> RusGasPipe[Turkmen-Russia gas pipeline]

U_Gas[Uzbek Gasfields] --> RusGasPipe

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end
subgraph Falklands
  SeaLion[Sea Lion oil] --> FPSO[FPSO (production)]
  FPSO --> Tankers[Shuttle tankers]
  Tankers --> UK_Port[UK/Europe via Atlantic]
end

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Commercial: Costs in Russia vary by project. Onshore West Siberian oil costs are relatively low (OPEX ~\$3–5/bbl). Deep offshore (Arctic, Sakhalin) capex is very high (\$40–70/bbl *breakeven*). Fiscal regime: CIT 20%; **MET** (Mineral Extraction Tax) and export duties apply to oil (progressive with distance/quality) and gas. Royalty (fixed %) was abolished; state takes via high export tax. Sanctions make financing/insurance difficult. **Priority:** Given current sanctions, new BP investment in Russia is highly constrained. Existing JV stakes (e.g. TNK-BP, Sakhalin II) are under Russian control. Rather, Russia is a mature backstop provider but not open to new Western capital.

Kazakhstan

Reserves: Kazakhstan's proven *crude oil* ≈ 30 bn bbl [19†L45-L52] [104†L147-L154] (mainly in Caspian/Tengiz/Buzachi) and *gas* ≈ 2.4 tcm (≈ 85 Tcf) [104†L147-L154]. Notable fields: **Tengiz** (~25.5 bn bbl in place [61†L160-L162]), **Kashagan** (~13 bn bbl recoverable [59†L169-L172]), **Karachaganak** (1.2 tcm gas, 1 Gt condensate) [64†L163-L171]. There are smaller onshore oil fields (Kumkol, Uzen, Zhanazhol) and a growing offshore/Aral basin potential.

Basins/Geology: Key basins include the *Pre-Caspian (Turgai)* with deep Paleozoic (Permian to Cambrian) carbonates (Tengiz, Karachaganak reefs) and overlying Mesozoic. The **Tengiz** reservoir is in Devonian/Carboniferous carbonates, overlain by thick Permian salt [70†L378-L387]. The *Caspian Sea* shelf hosts Kashagan (Carboniferous limestone reservoirs) [59†L157-L161]. *Finer detail:* source rocks include Paleozoic shale and Bazhenov-equivalents.

Drilling/History: Kazakhstan has a mature industry (since 1970s Soviet era) but still undeveloped frontier (deepwater Caspian). Tengiz was discovered 1979, producing 1993 [62†L25-L32]. Kashagan discovered 2000, came on-stream 2016 (after delays). Karachaganak discovered 1979, producing since 1984. Drilling strategies: Tengiz uses horizontal wells and gas reinjection (with planned SAGD for viscosity). Kashagan uses vertical (zero-rig) or floaters in shallow Caspian; future projects plan multi-lateral horizontal. Onshore fields are drilled conventionally; ongoing exploration in Caspian deeper horizons (Basement). Development wells now routinely 5,000–6,000 m (deep sandstones).

Hazards/Regulation: Environment is harsh steppe/tundra – cold in winter, remote sites. Earthquakes are minor. Political risk is moderate (stable government), but anti-corruption ranks are low. Profit-risk: high cost of sour gas treatment. Regulators (Kazakhstan *Ministry of Energy* and *NCOC-KazMunayGas-Chevron joint venture*, etc.) impose content rules (Kazakh workforce, procurement). In 2023–24 Kazakhstan overhauled hydrocarbon taxation: modern PSA contracts (Tengiz expansion) get stable royalty/CIT; mature fields pay royalty based on oil price (0–17%) [75†L193-L201].

Logistics & Export: Kazakhstan's crude mainly flows via the **CPC pipeline** (720 kbpd to Novorossiysk/Black Sea) and the Atyrau–Samara pipeline (to Russia). A smaller capacity East–West oil pipeline (trans-Kazakhstan) reaches the Caspian port of Kuryk. Gas is exported via the Central Asia–Center pipeline system north to Russia. A new *Kazakhstan-China* pipeline carries oil (~20 Mt/y) to Xinjiang. (For gas, a parallel China pipeline exists via Turkmenistan.) To supply the UK/Europe, Kazakh oil would travel via

Black Sea tankers or via Russian lines (no direct route). Gas to Europe would need reverse flows through Russia (unlikely given politics).

Commercial: Typical costs: Tengiz Phase 1 capex ~\$2 bn (for 100 kbpd capacity) ⇒ ~\$20k/bpd; OPEX ~\$10–15/bbl. Kashagan Phase 1 cost >\$50 bn (for 440 kbpd), very high breakeven. Break-even prices have been estimated at ~\$30–40 for major Kazakh fields. Fiscal: CIT 20%, royalty on oil (currently 5–17% sliding) and fixed share for gas projects. No export tariff on Tengiz/Kashagan (PSA terms). Priority: BP's legacy stake (17.5%) in Tengiz makes this high priority – expansion projects (Future Growth, Wellhead Pressure Mgmt) raise plateau output. Kashagan is high risk/high capex (BP holds no stake). Karachaganak (BP sold its 13.5%) is now 80% Shell/ENI. BP could consider exploration in adjacent onshore/deepwater plays, but main returns are on current JV fields.

Uzbekistan

Reserves: Uzbekistan's proven *oil* ≈594 mbbl, *gas* ≈65 Tcf 【104†L168-L174】. Explored reserves: ~178 Mt oil/condensate and ~2,240 bcm gas 【25†L81-L84】, but most is categorized probable. Production is ~63 kbpd oil (declining) and ~1.5 Tcf gas in 2023 【104†L168-L174】.

Basins/Geology: Main basins: **Bukhara-Khiva** (west Uzbekistan) – Mesozoic clastics (Jurassic sandstones) hosting large gas (Gazli) and oil (Shurtan, Jarkuduk). **Surkhandarya** (south) – Permian-Triassic traps (fields like Dauletabad on Turkmen border). **Ustyurt Plateau** (northwest) – Cretaceous-Permian gas fields (Shakhpakhty). **Fergana** (east) – Paleozoic/Neogene traps (small oil/gas fields). The petroleum system: Jurassic sandstones are main reservoirs; source rocks are Jurassic/Cretaceous marine shales. Tight carbonate reefs are minor.

Drilling/History: Fergana was the early oil center (discovered 1929); Surkhandarya (Haudag oil 1934, Gazli gas 1956). Bukhara-Khiva gas drilling began 1953 (first Gazli well) 【66†L131-L139】. Modern drilling uses jacking rigs in shallow wells (~1–2 km). Reserves are generally in small to medium fields (<100 Mt), so high-grading is needed. Exploration: 3D seismic since 2000s, targeting deep plays (Tarim-stage in Surkhandarya, Paleozoic in Ustyurt). Technology: acidizing/horizontal in carbonates, multi-layer perforating.

Hazards/Regulation: Landlocked terrain – very hot summers, minimal seismicity. Politically, moderate risk (stable but autocratic regime; some sanctions due to Afghan gas transit). Licensing via state body *Goskomgidromet*, with PSCs/Joint Ventures (e.g. with SOCAR, Total, LUKoil). BP was an early investor (Shurtan gas in 1997) but left in 2012 due to US secondary sanctions. Current focus is on gas, little recent foreign equity in onshore oil.

Logistics & Export: Uzbekistan's oil is tiny – mostly consumed domestically or piped to Kazakhstan/Russia. Gas exports: past via Central Asia–Center pipeline to Russia (but Russia cut off Uzbek gas in 2020). Modern pipelines: *Turkmenistan-Afghanistan-Pakistan-India* (TAPI) partly traverses Uzbek south; future plans include new links to China (via Turkmen base) and Afghanistan (Trans-Afghan). For UK/Europe export, Uzbek gas would need pipeline to Caspian or LNG: neither exists. Potential is via LNG through Iran or Pakistan (politically unlikely) or connecting to Turkmenexport systems (TAP/Nabucco – not realized).

Commercial: Costs are low by Western standards (\$5–10/bbl OPEX), but fields are small. Break-even price ~\$40–50. Uzbekistan reformed tax in 2018: PSCs have 8–20% royalty, CIT 20%. Foreign investors (SOCAR, Total) get 30-year PSCs. Domestic gas price is low; ex-growth markets are limited. BP exited, so

further BP projects seem unlikely unless sanctions lift. **Priority:** Marginal; perhaps interest in North Fergana deep drilling if returns high.

Turkmenistan

Reserves: Turkmenistan holds ~400 Tcf (≈ 11.3 tcm) proven *gas* 【104†L156-L164】 (5th globally) and minor oil (~600 mbbbl) 【104†L162-L165】. Main gas fields: **Galkynysh** (South Yoloten, Permian reef ~350 Tcf in place), **Dauletabad** (older large gas/condensate), **Shatlyk** (Jurassic gas), and Caspian Sea fields. Oil fields are small (Cheleken, Garagol) or uncommercial (Khetyler).

Basins/Geology: Predominantly **Amu-Darya Basin** – vast subsidence, thick Jurassic–Cretaceous clastics, Permian salt. Gas is in Permian–Triassic shelf carbonates and Cretaceous reservoirs. Source rocks: Paleozoic marine shales and Cretaceous coals. The **South Caspian Basin** (Gurgen, Dawletabad) has Jurassic carbonate reservoirs.

Drilling/History: First gas wells in 1940s; major development in 1950s. Post-independence, new drilling (e.g. onshore CPSs) and pipeline builds. Modern techniques: horizontal for tight gas, acid stimulation in carbonates. Foreign contractors (Petronas, Dragon Oil, ENI) are developing Galkynysh. Drilling is onshore and some shallow offshore. The wells can be deep (4–6 km) and often high-pressure.

Hazards/Regulation: Desert climate, extreme heat, dust. Politically, one-party state, risk of expropriation is low but contracts strict; some corruption. No sanctions (except Iran route issues). Licensing: state-controlled, production sharing. High flaring (inefficient gas capture) leads to pollution.

Logistics & Export: Gas exports: historically via the *Central Asia – Center* pipeline to Russia, and since 2009 via the *Turkmenistan–China* pipeline (90 Bcm/yr) 【104†L156-L164】. Plans for *Turkmenistan–Afghanistan–Pakistan–India (TAPI)* exist (gas to south). For oil: small pipeline Turkmenistan–Russia (Berezow, CPS), but mostly no oil exports. To reach UK/Europe, Turkmen gas would need a Trans-Caspian pipeline (Turkmen–Azerbaijan via seabed) into Turkey/EU or export LNG. EU projects (Nabucco, TAP) have stalled; Chinese deals lock output east. A major constraint is that Turkmen flared methane: high emissions could bar EU access.

Commercial: Very low operating cost, but also low export price: China pays modestly (around \$1.6–2/ Mmbtu). Capex: e.g. Galkynysh Field Development ~\$10–15 bbl OPEX (gas). No signature bonuses; companies accept government-set prices. Tax/CIT ~15–20%. State owns 100% gas. **Breakeven:** effectively domestic (near-zero) because markets limited. *Recommended:* Very unlikely for BP to enter (no open market, contracts state-to-state via Gazprom or CNPC). Turkmen is a gas exporter to China, not to UK.

Kyrgyzstan & Tajikistan

Both have only **modest hydrocarbon shows**. IEA reports Kyrgyz oil reserves ≈ 5 Mt (~37 mbbbl) and gas ≈ 6 bcm 【45†L74-L80】; Tajikistan ~0.6 Mt (≈ 4 mbbbl) oil and ~5.7 bcm gas 【51†L53-L60】 【52†L1-L4】. No commercial fields exist; primary resources are tectonic (Mesozoic subsalt plays in north Kyrgyzstan). No export infrastructure (landlocked, reliant on Russian imports). Political/technical risk is very high; these countries are **lowest priority** for exploration.

Falkland Islands (Islas Malvinas)

Reserves: The North Falklands Basin's Sea Lion oil field is estimated to contain ~917 mmbbl recoverable [75†L141-L149] . Other prospects ("Isobel", "Warrah", etc.) have only minor or undrilled yields so far. No gas fields confirmed.

Geological Setting: The Falklands sit on a conjugate margin of the South Atlantic, with Mesozoic rift basins [72†L133-L140] . Sea Lion lies in the **North Falkland Basin** on shallow continental shelf. Reservoirs are Upper Cretaceous sandstones (fluvial/deltaic); source is likely Permian shale (e.g. Beacon Supergroup analog). The system is analogous to west Africa basins. The basin is undersurveyed, but seismic indicates thick post-rift clastics. Other basins (South Falklands, Malvinas) are deeper water with potential but higher exploration risk.

Drilling History: Initial seismic (1970s–80s) showed Mesozoic basins [72†L133-L140] . Six wells were drilled by 2010; five hit oil/gas shows but none commercial. Sea Lion was discovered in 2010 by Desire/Rockhopper drilling (~450 m water). Shell and others had explored earlier but with dry results. (Sea Lion initial estimate 500 mmbbl, revised to ~900 mmbbl [75†L181-L189] .) No other discoveries of scale yet. Planned wells for forward development: Navitas/Rockhopper's Phase 1 calls for 11 subsea wells feeding an FPSO [81†L48-L52] .

Hazards/Regulation: Environmental conditions are severe: south Atlantic storms, cold, and sea ice in winter (though less at Sea Lion). Ecologically sensitive (whale migration, birds). Argentine sovereignty claim causes political risk (Argentina denounces any drilling [75†L169-L178]); UN allows UK jurisdiction, but diplomatic tension remains. Regulations follow strict UK/European safety and environmental standards (Falkland Government uses UK-like legislation [78†L63-L67]). All projects require full EIS (Navitas has published one). No issues with crew safety, but remote medevac is a concern (Stanley base is small).

Logistics & Export: Sea Lion will use a **Floating Production, Storage & Offtake (FPSO)** and shuttle tankers [81†L48-L52] . (No pipelines possible across oceans.) The FPSO (former Petrojarl 1 or similar) processes ~50 kbpd peak. Oil is then loaded onto Suezmax tankers in the Atlantic. Routes: likely via **Panama Canal** or around **Cape Horn** to reach Europe. E.g. 8,000+ km from Falklands to UK via Panama (~30–40 days transit). (No heavy chokepoints beyond canal; longer option is round Africa). Gas: none produced, but any gas would require re-injection or LNG (no gas export infrastructure).

Commercial: Development CAPEX is enormous per barrel: Sea Lion Phase 1 (170 mmbbl out of 917) costs ~US\$3–4 billion. OPEX is modest (FPSO hull ~\$5–7/bbl). Break-even has been reported around \$60–70/bbl (long lead logistics). Oil quality is ~30–35° API (comparable to Brent). Fiscal terms (Falklands): royalty 9% and corp tax 26% [79†L46-L54] . (No signature bonus; no state back-in.) NAVITAS' projections show ~£4 bn revenue on 170 mmbbl [75†L169-L177] . Environmental premiums may apply. **Risk:** Political (Argentina, insurance), and field risk (subsea complexity). **Priority:** Sea Lion Phase 1 is FID-approved (2025) with first oil 2028 [75†L165-L173] ; BP is not involved (Rockhopper/Navitas/Premier). For BP, only a new entrant playing amid high oil price and UK government support could be attractive, but competition and diplomacy make it marginal.

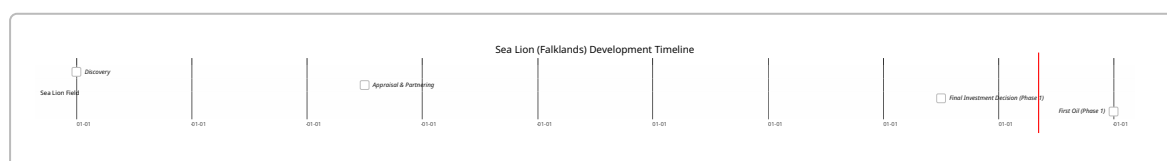


Table: Major Fields/Plays Comparison

Basin/Field (Major Country)	Hydrocarbons	Reserves (Proved/P+P)	Quality	Break- even / Cost	Key Hazards/ Risk
West Siberia (Russia)	Oil/Gas (Jurassic clastics)	~144 bn bbl oil, 1,300 Tcf gas discovered 【54†L233- L238】	Moderate (API 25–30)	~\$15– 25/bbl	Permafrost, aging wells/ pipelines, sanctions
Kashagan (Kazakhstan)	Oil (Caspian, Carbonates)	~13 bn bbl recoverable 【59†L169- L172】	Heavy, 17% S	~\$40– 60/bbl	Offshore H ₂ S, cost overruns, water depth
Tengiz (Kazakhstan)	Oil (Pre- Caspian, Carbonates)	25.5 bn bbl oil- in-place 【61†L160- L162】	Heavy, high pressure	~\$15– 25/bbl	Very high sulfur 【62†L24- L32】 , depth
Karachaganak (Kazakhstan)	Gas/Cond (Permian reef)	1.2 tcm gas, 1 Gt condensate 【64†L163- L171】	Rich condensate	~\$10– 15/bbl	Sour gas (H ₂ S 3–5%) 【64†L203- L211】
Bukhara-Khiva (Uzbekistan)	Oil/Gas (Jurassic)	~0.6 bn bbl oil, ~50 Tcf gas (all fields)	Lean oil, dry gas	~\$40– 50/bbl	Limited pipelines, low market price
Galkynysh (Turkmenistan)	Gas (Permian carbonates)	~400 Tcf gas 【104†L156- L164】	Dry, high GOR	<\$5/ Mcf	Methane emissions (EU restrictions), China-only market
Sea Lion (Falklands)	Oil (Cretaceous sand)	≈0.9 bn bbl recoverable 【75†L141- L149】	Light/ Medium	~\$60– 70/bbl	Remote offshore, South Atlantic storms, political dispute

Sources: Reserve and field data are from official/industry reports 【9†L57-L63】 【104†L147-L154】 【104†L156-L164】 【104†L168-L174】 . Geologic and field-specific information is from geoscientific literature and company disclosures 【54†L213-L221】 【59†L157-L161】 【64†L163-L171】 . Fiscal and regulatory details are from government publications 【79†L46-L54】 . Risk assessments are qualitative, based on climate, political and technical factors.